

PocketSmart – Startup and Main Function (Backend: FastAPI Integration)

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1. Topic

Writing the startup logic and main function for the FastAPI backend

Epic: Backend Development

2. Project Overview

PocketSmart is an AI-powered app that helps users track spending, set budgets, and get personalized money-saving suggestions. The startup and main function make sure the backend starts correctly and all its parts are ready before users send requests.

3. Objective

To write the startup process and the main function that prepare the database, load the configuration, connect the AI model, register the routes, and run the FastAPI application with Uvicorn.

4. Work Steps

- 1 Import the required packages, project modules, and environment settings.
- 2 Create the FastAPI application instance and set its title and version.
- 3 Write the startup function to load the configuration and API keys from the environment.
- 4 Create the database tables and check that the connection works at startup.
- 5 Initialize the AI model connection and confirm it is ready to use.
- 6 Register all routers for expenses, budgets, insights, and user management.
- 7 Add a shutdown function to close the database connection safely.
- 8 Add error handling so the application shows a clear message if startup fails.
- 9 Write the main function to run the app with Uvicorn and test that it starts properly.

5. Main Parts of the Startup and Main Function

- Imports and environment configuration
- FastAPI application initialization
- Startup event (database and AI model setup)
- Router registration
- Shutdown event and cleanup
- Main function and Uvicorn server run